CONCEPTUAL SERIES
ScyllaDB

Name: ____________ Cohort/Batch: ____________ Date: ____________ Score: ____

SCYLLADB PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is ScyllaDB and what are its core strengths? *Threat Model*

Q2 How does ScyllaDB guarantee consistency and handle transactions? *Architecture*

Q3 What index types does ScyllaDB support and when do you use each? *Configuration*

Q4 How do you design a schema or data model in ScyllaDB? *Hardening*

Q5 How does ScyllaDB scale horizontally? *SSO / IdP*

Q6 How do you monitor and tune slow queries in ScyllaDB? *Observability*

Q7 What backup and restore strategies does ScyllaDB support? *Lifecycle*

Q8 How do you handle schema migrations in ScyllaDB? *Compliance*

Q9 What security features does ScyllaDB provide? *Testing*

Q10 What are common anti-patterns when using ScyllaDB in production? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 ScyllaDB is a relational, document, key-value, graph, — Mitigation

ScyllaDB is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A2 ScyllaDB provides either full ACID transactions or — Primitives

ScyllaDB provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A3 ScyllaDB offers B-tree, hash, full-text, spatial, or — Hardening

ScyllaDB offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A4 Schema design in ScyllaDB involves normalizing relations — Gotchas

Schema design in ScyllaDB involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A5 ScyllaDB scales via replication (read replicas) for — Federation

ScyllaDB scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some ScyllaDB implementations handle this automatically; others require manual configuration.

A6 ScyllaDB provides a slow query log, explain/explain — Observability

ScyllaDB provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A7 ScyllaDB supports logical backups (export SQL or — Automation

ScyllaDB supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A8 Schema migrations in ScyllaDB are managed with — Governance

Schema migrations in ScyllaDB are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A9 ScyllaDB supports role-based access control, encrypted — Assessment

ScyllaDB supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A10 Common mistakes include selecting all columns instead — Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.